

## **TD 2 : Exploiter le code au moment de l'exécution**

**Objectifs pédagogiques :**

- Réflexivité
- Annotation
- Polymorphisme

### **Exercice 1 – Un toString pour tous**

Écrire une méthode statique `String toString(Object o)` générique (au sens général, pas au sens généricité Java) qui permet d'avoir une représentation d'un objet quelconque sous la forme d'une chaîne caractère. On doit respecter le format suivant

```
NomDeClasseReelle(typeatt1 name_att1=val_att1, typeatt2 name_att2=val_att2, ....)
```

On ne prendra pas en compte les attributs statiques.

### **Exercice 2 – Connaître ses types**

Écrire une méthode statique `getTypes` qui pour un objet `o` donné renvoie la liste des types de cet objet.

### **Exercice 3 – Annotation et introspection**

Nous souhaitons écrire une annotation `ExecuteOnBuilding` de méthode qui indique si une méthode doit être exécutée automatiquement après l'instanciation de l'objet. On pourra paramétrer l'annotation par un entier qui indiquera le nombre de fois que l'on souhaite exécuter la méthode (valeur par défaut = 1). On supposera que les méthodes ne prennent pas d'argument et ne retournent rien.

#### **Question 1**

Donner le code de l'annotation `ExecuteOnBuilding`

#### **Question 2**

Écrire une méthode statique `deploy` qui prend une classe en paramètre, qui instancie une nouvelle instance de cette classe, qui exécute les méthodes marquées par l'annotation `ExecuteOnBuilding` et qui renvoie un objet du type de la classe.

# Class Class<T>

- [java.lang.Object](#)
  - java.lang.Class<T>
- Type Parameters:

T - the type of the class modeled by this Class object. For example, the type of String.class is Class<String>. Use Class<?> if the class being modeled is unknown.

All Implemented Interfaces:

[Serializable](#), [AnnotatedElement](#), [GenericDeclaration](#), [Type](#)

```
public final class Class<T>
    extends Object
    implements Serializable, GenericDeclaration, Type, AnnotatedElement
```

Instances of the class Class represent classes and interfaces in a running Java application. An enum is a kind of class and an annotation is a kind of interface. Every array also belongs to a class that is reflected as a Class object that is shared by all arrays with the same element type and number of dimensions. The primitive Java types (boolean, byte, char, short, int, long, float, and double), and the keyword void are also represented as Class objects. Class has no public constructor. Instead Class objects are constructed automatically by the Java Virtual Machine as classes are loaded and by calls to the defineClass method in the class loader.

The following example uses a Class object to print the class name of an object:

```
void printClassName(Object obj) {
    System.out.println("The class of " + obj +
        " is " + obj.getClass().getName());
}
```

## Modifier and Type

```
<U> Class<? extends U>
    I
    boolean
    static Class<?>
    static Class<?>
    AnnotatedType[]
    AnnotatedType
    <A extends Annotation>
        A
    Annotation[]
    <A extends Annotation>
        A[]
    String
```

[asSubclass](#)([Class](#)<U> clazz)

Casts this Class object to represent a subclass of the class represented by the specified class object.

[cast](#)([Object](#) obj)

Casts an object to the class or interface represented by this Class object.

[desiredAssertionStatus](#)()

Returns the assertion status that would be assigned to this class if it were to be initialized at the time this method is invoked.

[forName](#)([String](#) className)

Returns the Class object associated with the class or interface with the given string name.

[forName](#)([String](#) name, boolean initialize, [ClassLoader](#) loader)

Returns the Class object associated with the class or interface with the given string name, using the given class loader.

[getAnnotatedInterfaces](#)()

Returns an array of [AnnotatedType](#) objects that represent the use of types to specify superinterfaces of the entity represented by this Class object.

[getAnnotatedSuperclass](#)()

Returns an [AnnotatedType](#) object that represents the use of a type to specify the superclass of the entity represented by this Class object.

[getAnnotation](#)([Class](#)<A> annotationClass)

Returns this element's annotation for the specified type if such an annotation is present, else null.

[getAnnotations](#)()

Returns annotations that are present on this element.

[getAnnotationsByType](#)([Class](#)<A> annotationClass)

Returns annotations that are associated with this element.

[getCanonicalName](#)()

Returns the canonical name of the underlying class as defined by the Java Language Specification.

[getClasses](#)()

Returns an array containing Class objects representing all the public classes and interfaces that are members of the class represented by this Class object.

[getClassLoader](#)()

Returns the class loader for the class.

[getComponentType](#)()

Returns the Class representing the component type of an array.

[getConstructor](#)([Class](#)<?>... parameterTypes)

Returns a Constructor object that reflects the specified public constructor of the class represented by this Class object.

[getConstructors](#)()

Returns an array containing Constructor objects reflecting all the public constructors of the class represented by this Class object.

[getDeclaredAnnotation](#)([Class](#)<A> annotationClass)

Returns this element's annotation for the specified type if such an annotation is directly present, else null.

[getDeclaredAnnotations](#)()

Returns annotations that are directly present on this element.

[getDeclaredAnnotationsByType](#)([Class](#)<A> annotationClass)

Returns this element's annotation(s) for the specified type if such annotations are either directly present or indirectly present.

[getDeclaredClasses](#)()

Returns an array of Class objects reflecting all the classes and interfaces declared as members of the class represented by this Class object.

[getDeclaredConstructor](#)([Class](#)<?>... parameterTypes)

Returns a Constructor object that reflects the specified constructor of the class or interface represented by this Class object.

[getDeclaredConstructors](#)()

Returns an array of Constructor objects reflecting all the constructors declared by the class represented by this Class object.

[getDeclaredField](#)([String](#) name)

Returns a Field object that reflects the specified declared field of the class or interface represented by this Class object.

[getDeclaredFields](#)()

Returns an array of Field objects reflecting all the fields declared by the class or interface represented by this Class object.

[getDeclaredMethod](#)([String](#) name, [Class](#)<?>... parameterTypes)

Returns a Method object that reflects the specified declared method of the class or interface represented by this Class object.

[getDeclaredMethods](#)()

Returns an array containing Method objects reflecting all the declared methods of the class or interface represented by this Class object, including public, protected, default (package) access, and private methods, but excluding inherited methods.

[getDeclaringClass](#)()

If the class or interface represented by this Class object is a member of another class, returns the Class object representing the class in which it was declared.

[getEnclosingClass](#)()

Returns the immediately enclosing class of the underlying class.

[getEnclosingConstructor](#)()

If this Class object represents a local or anonymous class within a constructor, returns a Constructor object representing the immediately enclosing constructor of the underlying class.

[getEnclosingMethod](#)()

If this Class object represents a local or anonymous class within a method, returns a Method object representing the immediately enclosing method of the underlying class.

[getEnumConstants](#)()

Returns the elements of this enum class or null if this Class object does not represent an enum type.

[getField](#)([String](#) name)

Returns a Field object that reflects the specified public member field of the class or interface

|                                                   |                                                                                                                                                                                                                                                                                                                               |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | represented by this <code>Class</code> object.                                                                                                                                                                                                                                                                                |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>Field[]</code>                              | <a href="#"><code>getFieldIds()</code></a><br>Returns an array containing <code>Field</code> objects reflecting all the accessible public fields of the class or interface represented by this <code>Class</code> object.                                                                                                     | boolean             | <a href="#"><code>isInstance(Object obj)</code></a><br>Determines if the specified <code>Object</code> is assignment-compatible with the object represented by this <code>Class</code> .                                                                                                                                                                                                                                                                                                                   |
| <code>Type[]</code>                               | <a href="#"><code>getGenericInterfaces()</code></a><br>Returns the <code>Types</code> representing the interfaces directly implemented by the class or interface represented by this object.                                                                                                                                  | boolean             | <a href="#"><code>isInterface()</code></a><br>Determines if the specified <code>Class</code> object represents an interface type.                                                                                                                                                                                                                                                                                                                                                                          |
| <code>Type</code>                                 | <a href="#"><code>getGenericSuperclass()</code></a><br>Returns the <code>Type</code> representing the direct superclass of the entity (class, interface, primitive type or void) represented by this <code>Class</code> .                                                                                                     | boolean             | <a href="#"><code>isLocalClass()</code></a><br>Returns <code>true</code> if and only if the underlying class is a local class.                                                                                                                                                                                                                                                                                                                                                                             |
| <code>Class&lt;?&gt;[]</code>                     | <a href="#"><code>getInterfaces()</code></a><br>Determines the interfaces implemented by the class or interface represented by this object.                                                                                                                                                                                   | boolean             | <a href="#"><code>isMemberClass()</code></a><br>Returns <code>true</code> if and only if the underlying class is a member class.                                                                                                                                                                                                                                                                                                                                                                           |
| <code>Method</code>                               | <a href="#"><code>getMethod(String name, Class&lt;?&gt;... parameterTypes)</code></a><br>Returns a <code>Method</code> object that reflects the specified public member method of the class or interface represented by this <code>Class</code> object.                                                                       | boolean             | <a href="#"><code>isPrimitive()</code></a><br>Determines if the specified <code>Class</code> object represents a primitive type.                                                                                                                                                                                                                                                                                                                                                                           |
| <code>Method[]</code>                             | <a href="#"><code>getMethods()</code></a><br>Returns an array containing <code>Method</code> objects reflecting all the public methods of the class or interface represented by this <code>Class</code> object, including those declared by the class or interface and those inherited from superclasses and superinterfaces. | <code>String</code> | <a href="#"><code>isSynthetic()</code></a><br>Returns <code>true</code> if this class is a synthetic class; returns <code>false</code> otherwise.                                                                                                                                                                                                                                                                                                                                                          |
| <code>int</code>                                  | <a href="#"><code>getModifiers()</code></a><br>Returns the Java language modifiers for this class or interface, encoded in an integer.                                                                                                                                                                                        | <code>String</code> | <a href="#"><code>newInstance()</code></a><br>Creates a new instance of the class represented by this <code>Class</code> object.                                                                                                                                                                                                                                                                                                                                                                           |
| <code>String</code>                               | <a href="#"><code>getName()</code></a><br>Returns the name of the entity (class, interface, array class, primitive type, or void) represented by this <code>Class</code> object, as a <code>String</code> .                                                                                                                   |                     | <a href="#"><code>toGenericString()</code></a><br>Returns a string describing this <code>Class</code> , including information about modifiers and type parameters.                                                                                                                                                                                                                                                                                                                                         |
| <code>Package</code>                              | <a href="#"><code>getPackage()</code></a><br>Gets the package for this class.                                                                                                                                                                                                                                                 |                     | <a href="#"><code>toString()</code></a><br>Converts the object to a string.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>ProtectionDomain</code>                     | <a href="#"><code>getProtectionDomain()</code></a><br>Returns the <code>ProtectionDomain</code> of this class.                                                                                                                                                                                                                |                     | <ul style="list-style-type: none"> <li><b>Methods inherited from class <code>java.lang.Object</code></b><br/><a href="#"><code>clone</code></a>, <a href="#"><code>equals</code></a>, <a href="#"><code>finalize</code></a>, <a href="#"><code>getClass</code></a>, <a href="#"><code>hashCode</code></a>, <a href="#"><code>notify</code></a>, <a href="#"><code>notifyAll</code></a>, <a href="#"><code>wait</code></a>, <a href="#"><code>wait</code></a>, <a href="#"><code>wait</code></a></li> </ul> |
| <code>URL</code>                                  | <a href="#"><code>getResource(String name)</code></a><br>Finds a resource with a given name.                                                                                                                                                                                                                                  |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>InputStream</code>                          | <a href="#"><code>getResourceAsStream(String name)</code></a><br>Finds a resource with a given name.                                                                                                                                                                                                                          |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>Object[]</code>                             | <a href="#"><code>getSigners()</code></a><br>Gets the signers of this class.                                                                                                                                                                                                                                                  |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>String</code>                               | <a href="#"><code>getSimpleName()</code></a><br>Returns the simple name of the underlying class as given in the source code.                                                                                                                                                                                                  |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>Class&lt;? super I&gt;</code>               | <a href="#"><code>getSuperclass()</code></a><br>Returns the <code>Class</code> representing the superclass of the entity (class, interface, primitive type or void) represented by this <code>Class</code> .                                                                                                                  |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>String</code>                               | <a href="#"><code>getTypeName()</code></a><br>Return an informative string for the name of this type.                                                                                                                                                                                                                         |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>TypeVariable&lt;Class&lt;T&gt;&gt;[]</code> | <a href="#"><code>getTypeParameters()</code></a><br>Returns an array of <code>TypeVariable</code> objects that represent the type variables declared by the generic declaration represented by this <code>GenericDeclaration</code> object, in declaration order.                                                             |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| boolean                                           | <a href="#"><code>isAnnotation()</code></a><br>Returns <code>true</code> if this <code>Class</code> object represents an annotation type.                                                                                                                                                                                     | boolean             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| boolean                                           | <a href="#"><code>isAnnotationPresent(Class&lt;? extends Annotation&gt; annotationClass)</code></a><br>Returns <code>true</code> if an annotation for the specified type is <i>present</i> on this element, else <code>false</code> .                                                                                         | boolean             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| boolean                                           | <a href="#"><code>isAnonymousClass()</code></a><br>Returns <code>true</code> if and only if the underlying class is an anonymous class.                                                                                                                                                                                       | boolean             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| boolean                                           | <a href="#"><code>isArray()</code></a><br>Determines if this <code>Class</code> object represents an array class.                                                                                                                                                                                                             | boolean             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| boolean                                           | <a href="#"><code>isAssignableFrom(Class&lt;?&gt; cls)</code></a><br>Determines if the class or interface represented by this <code>Class</code> object is either the same as, or is a superclass or superinterface of, the class or interface represented by the specified <code>Class</code> parameter.                     | boolean             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| boolean                                           | <a href="#"><code>isEnum()</code></a><br>Returns <code>true</code> if and only if this class was declared as an enum in the source code.                                                                                                                                                                                      | boolean             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

java.lang.reflect

## Class Field

- [java.lang.Object](#)
  - [java.lang.reflect.AccessibleObject](#)
- All Implemented Interfaces:
  - [AnnotatedElement](#), [Member](#)

```
public final class Field
extends AccessibleObject
implements Member
```

A `Field` provides information about, and dynamic access to, a single field of a class or an interface. The reflected field may be a class (static) field or an instance field. A `Field` permits widening conversions to occur during a get or set access operation, but throws an `IllegalArgumentException` if a narrowing conversion would occur.

| Modifier and Type                           | Method and Description                                                                                                                                                                                            |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| boolean                                     | <a href="#">equals(Object obj)</a><br>Compares this <code>Field</code> against the specified object.                                                                                                              |
| <a href="#">Object</a>                      | <a href="#">get(Object obj)</a><br>Returns the value of the field represented by this <code>Field</code> , on the specified object.                                                                               |
| <a href="#">AnnotatedType</a>               | <a href="#">getAnnotatedType()</a><br>Returns an <code>AnnotatedType</code> object that represents the use of a type to specify the declared type of the field represented by this <code>Field</code> .           |
| <T extends <a href="#">Annotation</a> > T   | <a href="#">getAnnotation(Class&lt;T&gt; annotationClass)</a><br>Returns this element's annotation for the specified type if such an annotation is <i>present</i> , else null.                                    |
| <T extends <a href="#">Annotation</a> > T[] | <a href="#">getAnnotationsByType(Class&lt;T&gt; annotationClass)</a><br>Returns annotations that are <i>associated</i> with this element.                                                                         |
| boolean                                     | <a href="#">getBoolean(Object obj)</a><br>Gets the value of a static or instance <code>boolean</code> field.                                                                                                      |
| byte                                        | <a href="#">getBytes(Object obj)</a><br>Gets the value of a static or instance <code>byte</code> field.                                                                                                           |
| char                                        | <a href="#">getChar(Object obj)</a><br>Gets the value of a static or instance field of type <code>char</code> or of another primitive type convertible to type <code>char</code> via a widening conversion.       |
| <a href="#">Annotation[]</a>                | <a href="#">getDeclaredAnnotations()</a><br>Returns annotations that are <i>directly present</i> on this element.                                                                                                 |
| <a href="#">Class</a> <?>                   | <a href="#">getDeclaringClass()</a><br>Returns the <code>Class</code> object representing the class or interface that declares the field represented by this <code>Field</code> object.                           |
| double                                      | <a href="#">getDouble(Object obj)</a><br>Gets the value of a static or instance field of type <code>double</code> or of another primitive type convertible to type <code>double</code> via a widening conversion. |
| float                                       | <a href="#">getFloat(Object obj)</a><br>Gets the value of a static or instance field of type <code>float</code> or of another primitive type convertible to type <code>float</code> via a widening conversion.    |
| <a href="#">Type</a>                        | <a href="#">getGenericType()</a><br>Returns a <code>Type</code> object that represents the declared type for the field represented by this <code>Field</code> object.                                             |
| int                                         | <a href="#">getInt(Object obj)</a><br>Gets the value of a static or instance field of type <code>int</code> or of another primitive type convertible to type <code>int</code> via a widening conversion.          |
| long                                        | <a href="#">getLong(Object obj)</a><br>Gets the value of a static or instance field of type <code>long</code> or of another primitive type convertible to type <code>long</code> via a widening conversion.       |

[getModifiers\(\)](#)  
Returns the Java language modifiers for the field represented by this `Field` object, as an integer.

[getName\(\)](#)  
Returns the name of the field represented by this `Field` object.

[getShort\(Object obj\)](#)  
Gets the value of a static or instance field of type `short` or of another primitive type convertible to type `short` via a widening conversion.

[getType\(\)](#)  
Returns a `Class` object that identifies the declared type for the field represented by this `Field` object.

[hashCode\(\)](#)  
Returns a hashcode for this `Field`.

[isEnumConstant\(\)](#)  
Returns `true` if this field represents an element of an enumerated type; returns `false` otherwise.

[isSynthetic\(\)](#)  
Returns `true` if this field is a synthetic field; returns `false` otherwise.

[set\(Object obj, Object value\)](#)  
Sets the field represented by this `Field` object on the specified object argument to the specified new value.

[setBoolean\(Object obj, boolean z\)](#)  
Sets the value of a field as a `boolean` on the specified object.

[setByte\(Object obj, byte b\)](#)  
Sets the value of a field as a `byte` on the specified object.

[setChar\(Object obj, char c\)](#)  
Sets the value of a field as a `char` on the specified object.

[setDouble\(Object obj, double d\)](#)  
Sets the value of a field as a `double` on the specified object.

[setFloat\(Object obj, float f\)](#)  
Sets the value of a field as a `float` on the specified object.

[setInt\(Object obj, int i\)](#)  
Sets the value of a field as an `int` on the specified object.

[setLong\(Object obj, long l\)](#)  
Sets the value of a field as a `long` on the specified object.

[setShort\(Object obj, short s\)](#)  
Sets the value of a field as a `short` on the specified object.

[toGenericString\(\)](#)  
Returns a string describing this `Field`, including its generic type.

[toString\(\)](#)  
Returns a string describing this `Field`.

- Methods inherited from class [java.lang.reflect.AccessibleObject](#)**  
[getAnnotations](#), [getDeclaredAnnotation](#), [getDeclaredAnnotationsByType](#), [isAccessible](#), [isAnnotationPresent](#), [setAccessible](#), [setAccessible](#)
- Methods inherited from class [java.lang.Object](#)**  
[clone](#), [finalize](#), [getClass](#), [notify](#), [notifyAll](#), [wait](#), [wait](#)

# Class Method

- [java.lang.Object](#)
  - [java.lang.reflect.AccessibleObject](#)
  - [java.lang.reflect.Executable](#)
  - [java.lang.reflect.Method](#)
- All Implemented Interfaces:
  - [AnnotatedElement](#), [GenericDeclaration](#), [Member](#)

## Modifier and Type

boolean

## Method and Description

[equals\(Object obj\)](#)  
Compares this Method against the specified object.

[getAnnotatedReturnType\(\)](#)

Returns an AnnotatedType object that represents the use of a type to specify the return type of the method/constructor represented by this Executable.

[getAnnotation\(Class<T> annotationClass\)](#)

Returns this element's annotation for the specified type if such an annotation is *present*, else null.

[getDeclaredAnnotations\(\)](#)

Returns annotations that are *directly present* on this element.

[getDeclaringClass\(\)](#)

Returns the Class object representing the class or interface that declares the executable represented by this object.

[getDefaultValue\(\)](#)

Returns the default value for the annotation member represented by this Method instance.

[getExceptionTypes\(\)](#)

Returns an array of Class objects that represent the types of exceptions declared to be thrown by the underlying executable represented by this object.

[getGenericExceptionTypes\(\)](#)

Returns an array of Type objects that represent the exceptions declared to be thrown by this executable object.

[getGenericParameterTypes\(\)](#)

Returns an array of Type objects that represent the formal parameter types, in declaration order, of the executable represented by this object.

[getGenericReturnType\(\)](#)

Returns a Type object that represents the formal return type of the method represented by this Method object.

[getModifiers\(\)](#)

Returns the Java language [modifiers](#) for the executable represented by this object.

[getName\(\)](#)

Returns the name of the method represented by this Method object, as a String.

[getParameterAnnotations\(\)](#)

Returns an array of arrays of Annotations that represent the annotations on the formal parameters, in declaration order, of the Executable represented by this object.

[getParameterCount\(\)](#)

Returns the number of formal parameters (whether explicitly declared or implicitly declared or neither) for the executable represented by this object.

[getParameterTypes\(\)](#)

Returns an array of Class objects that represent the formal parameter types, in declaration order, of the executable represented by this object.

[getReturnType\(\)](#)

Returns a Class object that represents the formal return type of the method represented by this Method object.

[TypeVariable<Method>\[\]](#)

[getTypeParameters\(\)](#)

Returns an array of TypeVariable objects that represent the type variables declared by the generic declaration represented by this GenericDeclaration object, in declaration order.

[hashCode\(\)](#)

Returns a hashcode for this Method.

[invoke\(Object obj, Object... args\)](#)

Invokes the underlying method represented by this Method object, on the specified object with the specified parameters.

[isBridge\(\)](#)

Returns true if this method is a bridge method; returns false otherwise.

[isDefault\(\)](#)

Returns true if this method is a default method; returns false otherwise.

[isSynthetic\(\)](#)

Returns true if this executable is a synthetic construct; returns false otherwise.

[isVarArgs\(\)](#)

Returns true if this executable was declared to take a variable number of arguments; returns false otherwise.

[toGenericString\(\)](#)

Returns a string describing this Method, including type parameters.

[toString\(\)](#)

Returns a string describing this Method.

- **Methods inherited from class java.lang.reflect.Executable**

[getAnnotatedExceptionTypes](#), [getAnnotatedParameterTypes](#), [getAnnotatedReceiverType](#), [getAnnotationsByType](#), [getParameters](#)

- **Methods inherited from class java.lang.reflect.AccessibleObject**

[getAnnotations](#), [getDeclaredAnnotation](#), [getDeclaredAnnotationsByType](#), [isAccessible](#), [isAnnotationPresent](#), [setAccessible](#), [setAccessible](#)

- **Methods inherited from class java.lang.Object**

[clone](#), [finalize](#), [getClass](#), [notify](#), [notifyAll](#), [wait](#), [wait](#)

- **Methods inherited from interface java.lang.reflect.AnnotatedElement**

[getAnnotations](#), [getDeclaredAnnotation](#), [getDeclaredAnnotationsByType](#), [isAnnotationPresent](#)

# Class Constructor<T>

- [java.lang.Object](#)
    - [java.lang.reflect.AccessibleObject](#)
    - [java.lang.reflect.Executable](#)
    - [java.lang.reflect.Constructor<T>](#)
  - Type Parameters:
    - T - the class in which the constructor is declared
- All Implemented Interfaces:
- [AnnotatedElement](#), [GenericDeclaration](#), [Member](#)

## Modifier and Type

boolean

[AnnotatedType](#)

[AnnotatedType](#)

<T extends [Annotation](#)>  
T

[Annotation\[\]](#)

[Class<T>](#)

[Class<?>\[\]](#)

[Type\[\]](#)

[Type\[\]](#)

int

[String](#)

[Annotation\[\]\[\]](#)

int

[Class<?>\[\]](#)

[TypeVariable<Constructor<T>>\[\]](#)

int

## Method and Description

[equals\(Object obj\)](#)

Compares this [Constructor](#) against the specified object.

[getAnnotatedReceiverType\(\)](#)

Returns an [AnnotatedType](#) object that represents the use of a type to specify the receiver type of the method/constructor represented by this [Executable](#) object.

[getAnnotatedReturnType\(\)](#)

Returns an [AnnotatedType](#) object that represents the use of a type to specify the return type of the method/constructor represented by this [Executable](#).

[getAnnotation\(Class<T> annotationClass\)](#)

Returns this element's annotation for the specified type if such an annotation is *present*, else null.

[getDeclaredAnnotations\(\)](#)

Returns annotations that are *directly present* on this element.

[getDeclaringClass\(\)](#)

Returns the [Class](#) object representing the class or interface that declares the executable represented by this object.

[getExceptionTypes\(\)](#)

Returns an array of [Class](#) objects that represent the types of exceptions declared to be thrown by the underlying executable represented by this object.

[getGenericExceptionTypes\(\)](#)

Returns an array of [Type](#) objects that represent the exceptions declared to be thrown by this executable object.

[getGenericParameterTypes\(\)](#)

Returns an array of [Type](#) objects that represent the formal parameter types, in declaration order, of the executable represented by this object.

[getModifiers\(\)](#)

Returns the Java language [modifiers](#) for the executable represented by this object.

[getName\(\)](#)

Returns the name of this constructor, as a string.

[getParameterAnnotations\(\)](#)

Returns an array of arrays of [Annotations](#) that represent the annotations on the formal parameters, in declaration order, of the [Executable](#) represented by this object.

[getParameterCount\(\)](#)

Returns the number of formal parameters (whether explicitly declared or implicitly declared or neither) for the executable represented by this object.

[getParameterTypes\(\)](#)

Returns an array of [Class](#) objects that represent the formal parameter types, in declaration order, of the executable represented by this object.

[getTypeParameters\(\)](#)

Returns an array of [TypeVariable](#) objects that represent the type variables declared by the generic declaration represented by this [GenericDeclaration](#) object, in declaration order.

[hashCode\(\)](#)

Returns a hashcode for this [Constructor](#).

boolean

boolean

I

[String](#)

[String](#)

- **Methods inherited from class [java.lang.reflect.Executable](#)**

[getAnnotatedExceptionTypes](#), [getAnnotatedParameterTypes](#), [getAnnotationsByType](#), [getParameters](#)

- **Methods inherited from class [java.lang.reflect.AccessibleObject](#)**

[getAnnotations](#), [getDeclaredAnnotation](#), [getDeclaredAnnotationsByType](#), [isAccessible](#), [isAnnotationPresent](#), [setAccessible](#), [setAccessible](#)

- **Methods inherited from class [java.lang.Object](#)**

[clone](#), [finalize](#), [getClass](#), [notify](#), [notifyAll](#), [wait](#), [wait](#)

- **Methods inherited from interface [java.lang.reflect.AnnotatedElement](#)**

[getAnnotations](#), [getDeclaredAnnotation](#), [getDeclaredAnnotationsByType](#), [isAnnotationPresent](#)

[isSynthetic\(\)](#)

Returns `true` if this executable is a synthetic construct; returns `false` otherwise.

[isVarArgs\(\)](#)

Returns `true` if this executable was declared to take a variable number of arguments; returns `false` otherwise.

[newInstance\(Object... initargs\)](#)

Uses the constructor represented by this [Constructor](#) object to create and initialize a new instance of the constructor's declaring class, with the specified initialization parameters.

[toGenericString\(\)](#)

Returns a string describing this [Constructor](#), including type parameters.

[toString\(\)](#)

Returns a string describing this [Constructor](#).